Lighthouse

AI agent orchestration & decision intelligence

DEMO & USER GUIDE

Four agents. Four owners. One decision.

How to open, use and present Lighthouse: the governance layer that reconciles a Sales, Finance, Operations and Legal AI agent for a 50-store Walmart U.S. pilot. Each agent is owned by a department lead, and a named executive makes the final call.

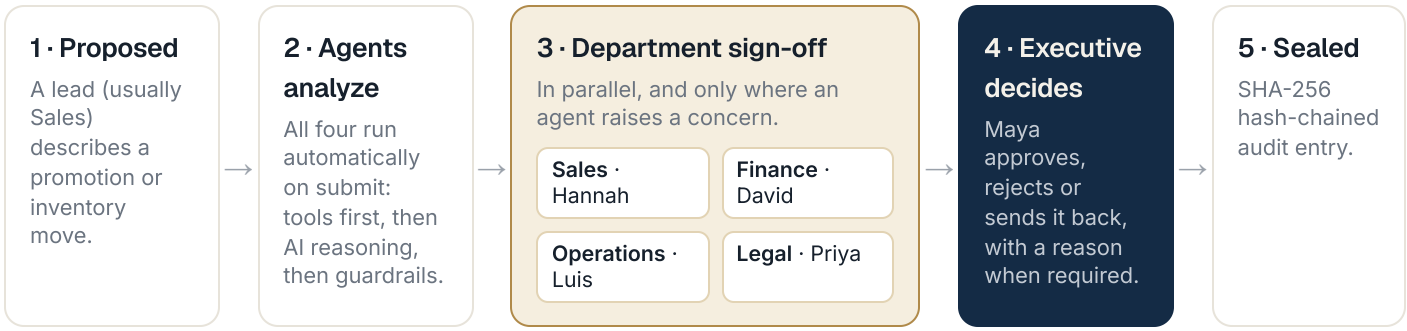
Live app	`https://lighthouse-roan-two.vercel.app`
Code	`github.com/BhavyaAk25/lighthouse`
Sign in	Pick a person. No passwords. The administrator uses PIN `4210`.
Cost	$0: Vercel Hobby, Neon free tier, Gemini and Groq free tiers.

ISTM 6210 / 4210 Capstone · Fall 2026 · The George Washington University
Ajay · Bhavya · Ehson · Jaehoon · Kunal · Wilfred · Faculty advisor: Dr. Soheil Sibdari
Academic prototype, not affiliated with Walmart Inc. Demand data is public Kaggle Walmart data; everything generated is labelled.

1 · THE IDEA

Lighthouse in one minute

Companies now run several specialised AI agents. Each can be right on its own and still disagree with the others: Sales sees demand, Finance sees a loss, Operations sees a full distribution center, Legal sees a risky claim. **Lighthouse is the layer above the agents.** Every agent answers in the same format with cited evidence. The department that owns the agent signs off on it, Lighthouse reconciles the four positions, and the executive decides. Every step lands in a tamper-evident log.



2 · THE PEOPLE

Who does what

PERSON	ROLE	DOES	CANNOT
MC **Maya Chen**	Executive · decision owner	Sees the full packet and all four sign-offs. Approves, rejects or sends back. Can overrule a Sales, Finance or Ops block with a reason, and can decide when a sign-off is overdue.	Overrule Legal; approve before required sign-offs
HB **Hannah Brooks**	Sales lead	Proposes promotions (submitting runs all four agents). Owns the Sales agent: demand, pricing, channel.	Sign for other departments
DO **David Okafor**	Finance lead	Owns the Finance agent: margin, ROI, break-even. Confirms, adjusts or blocks its answer.	Decide the case
LM **Luis Moreno**	Operations lead	Owns the Operations agent: inventory, DC capacity, fulfillment. Also proposes inventory moves.	Decide the case
PR **Priya Raman**	Legal lead	Owns the Legal agent: claims, privacy, state rules. Approves, approves with conditions, or rejects. **Her rejection is final.**	Approve a case a rule prohibits
JE **Jordan Ellis** `PIN 4210`	System administrator	Edits rules, thresholds and the Live AI / Replay setting (each change needs a reason). Resets the demo, tests providers, runs the test lab.	Sign off or decide (separation of duties)

3 · THE RULES

Why it is built this way, and the rules

Ownership

Each department lead owns their agent. They can re-run it, confirm it, change its answer with a reason, or block it. Nobody runs another department's agent.

Management by exception

A lead only signs when their agent says hold or reject, needs a review, is under 65% confident, or lacks evidence. In the 12 demo cases, about two out of three lanes clear automatically.

Accountability

The packet shows what the AI said and what each person signed. The executive can overrule Sales, Finance or Operations with a written reason. Legal can never be overruled.

A lead must sign when their agent…

- says Hold or Reject
- says Review required (for example, Legal flags a claim)
- is less than **65%** confident (admin can change this)
- is missing evidence (the output is marked invalid, never imputed)

Otherwise the lane shows *Cleared automatically*. A lead can still step in on any lane.

What a lead can do

- **Confirm** the agent's answer (a note is optional)
- **Adjust** it to proceed / with conditions / hold: needs a 20+ character reason, and lowers Lighthouse's confidence for that agent by 15%
- **Block** it: needs a reason. Legal's version is *Reject*.
- **Re-run my agent** at any time. Its old sign-off is cleared and must be given again.

When the executive can approve

- Every lane is cleared or signed, **and**
- Legal has not rejected it and no legal rule prohibits it
- A Sales, Finance or Ops **block** can be overruled with a written reason, and is recorded as an override
- Going against Lighthouse's recommendation, or a high-risk case, also needs a reason

Reject and Send back are allowed any time after the agents run.

Deadlines and escalation

- Each sign-off is due **48 hours** after the agents run (admin can change this)
- An overdue Sales, Finance or Ops lane escalates: the executive may decide without it, with a reason, and the record names who didn't sign
- An overdue **Legal** lane never escalates. Approval stays locked.

Lighthouse reconciles twice: first on the four AI answers, then again on what the departments actually signed. When the signed result differs, the decision brief says so in plain words: *"The agents alone said hold. After the departments signed off, Lighthouse recommends proceed with conditions."*

4 · GETTING READY

Before class (10 minutes)

- [ ] Open the live app and sign in as anyone. Click your name (top right): this menu switches person in one click and stays on the same page.
- [ ] In that menu, under *Presenter*, click **Reset demo cases**. Any time before class is fine: the demo clock stops 12 hours after a reset, so deadlines never all go overdue.
- [ ] *Rules & agents* → *Agents & AI* → *Test providers*. Both Gemini and Groq should show OK.
- [ ] In the same menu, choose **Live AI** if the network is good, or **Replay** if it isn't (saved AI answers, instant, identical results). The setting applies to everyone in that browser.
- [ ] Tip: give each teammate one persona and pass the laptop, or switch from the name menu. After every sign-off or decision a *Next for you* card points to the next case, and each desk opens with *Since you last looked*, so the audience sees every hand-off.

5 · THE DEMO

Demo script (about 8 minutes)

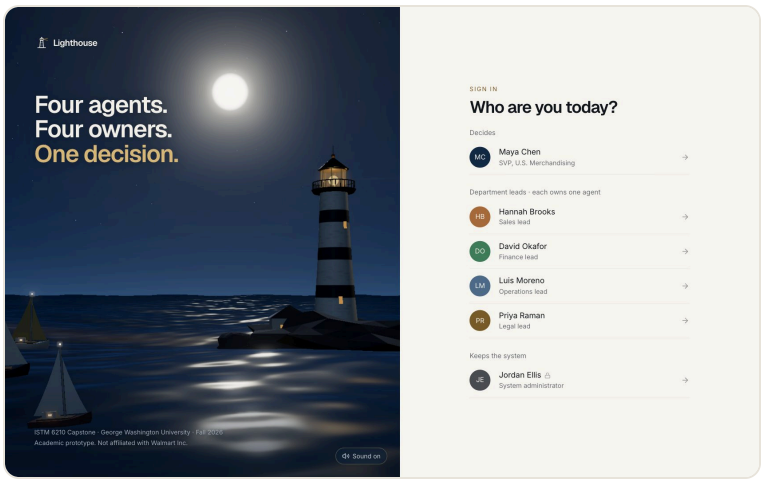
One promotion from proposal to sealed decision, with a detour where a person disagrees with their AI. Switch person from the **name menu** (top right): it stays on the same page, so every hand-off is one click.

TIME	PERSON	WHAT THE AUDIENCE SEES
0:00–0:45	Sign-in	Click the lighthouse: it lights up and the beams sweep (sound on). The five roles.
0:45–2:00	Hannah · Sales	Submits D-2026-001; four agents run live; she stays on the full analysis of her proposal
2:00–2:45	Priya · Legal	Same page, one click: only Legal is needed; she approves the claim with conditions
2:45–4:00	David · Finance	His desk shows what just happened; on D-2026-003 he re-runs his agent, then changes its "reject" with a reason
4:00–5:45	Maya · Executive	Previews on her desk; "AI alone said hold, sign-offs changed it"; an overdue sign-off; a locked case; approves D-2026-001
5:45–7:00	Optional loop	Maya sends D-2026-003 back → Hannah revises → David blocks → Maya overrules with a reason
7:00–8:00	Maya, Jordan	Flow, Dashboard, audit chain check, 18 acceptance tests

1 **Set the scene.** SIGN-IN SCREEN

Click the lighthouse on the left: the lamp ignites, two beams sweep the sea, and you hear the surf and the switch (mute is bottom right). Then point at the five people.

> *"A promotion touches four departments, and each now has an AI agent. In Lighthouse every agent has an owner, a person signs only when their agent has a concern, and a named executive makes the final call."*

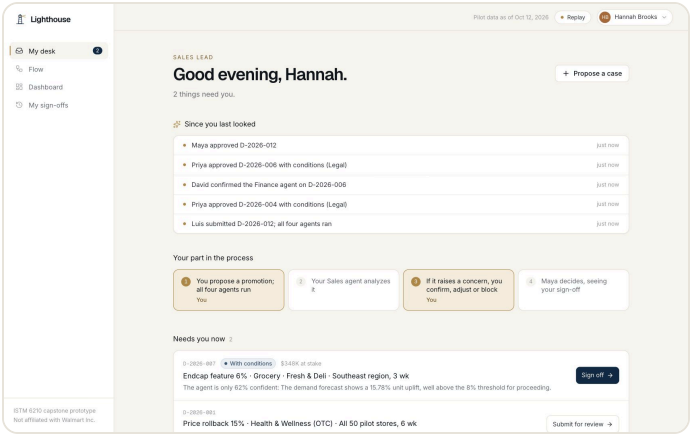


Sign-in: the lit lighthouse, and people grouped by what they do.

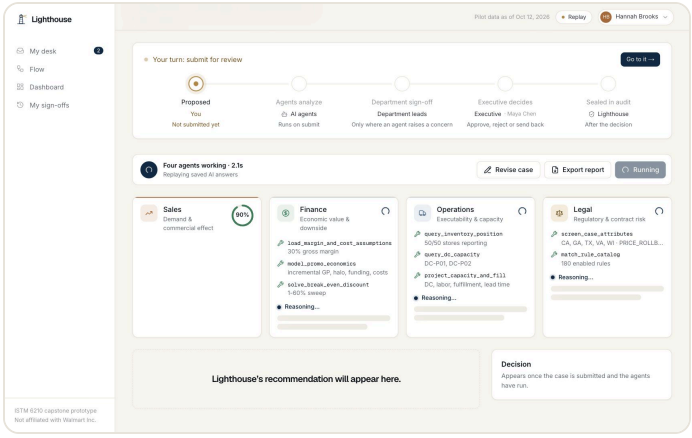
2 **Hannah proposes and submits.** HANNAH · SALES LEAD

Sign in as Hannah. Her desk shows two things: a sign-off on D-2026-007 and her draft D-2026-001 (15% OTC rollback, all 50 stores, "Clinically proven cold & flu relief"). Click **Submit for review** on the row, then **Submit for review** on the case.

> *"Submitting runs all four agents in parallel: tools first (you can see their names), then the model reasons over that evidence, then guardrails check it."*

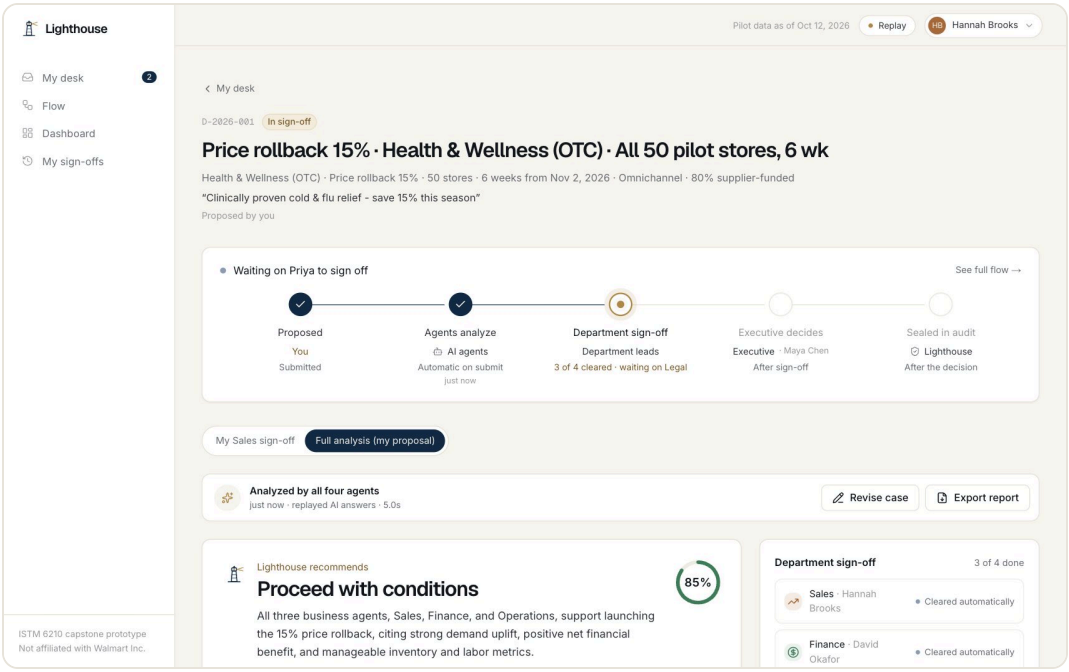


Hannah's desk: only her work, each with the reason it needs her.



Four agents mid-run, tool calls first, then reasoning.

When it finishes she stays on **Full analysis (my proposal)**. The top of the page is the **decision brief**: what Lighthouse recommends, in one line, and only what matters. The sign-off panel shows three departments cleared automatically, and the journey bar says *Waiting on Priya to sign off*.

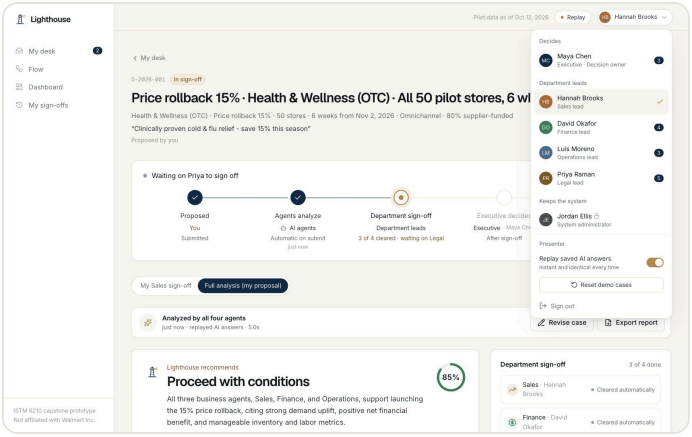


After submitting: the journey bar, the decision brief and the four departments' status.

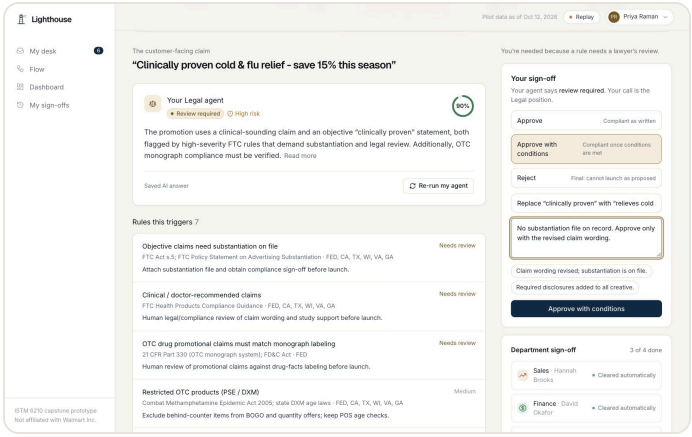
3 Priya signs off on the claim. PRIYA · LEGAL LEAD

Open the name menu and pick Priya. The page stays on D-2026-001 and now shows *her* lane: the claim, her agent's finding and 7 triggered rules (FTC substantiation, OTC monograph). Choose **Approve with conditions**, add the condition and a note, and sign.

> *"Legal is the only department needed here. Priya doesn't review the other three, they don't wait on her, and her rejection would be final: even the executive can't overrule it."*

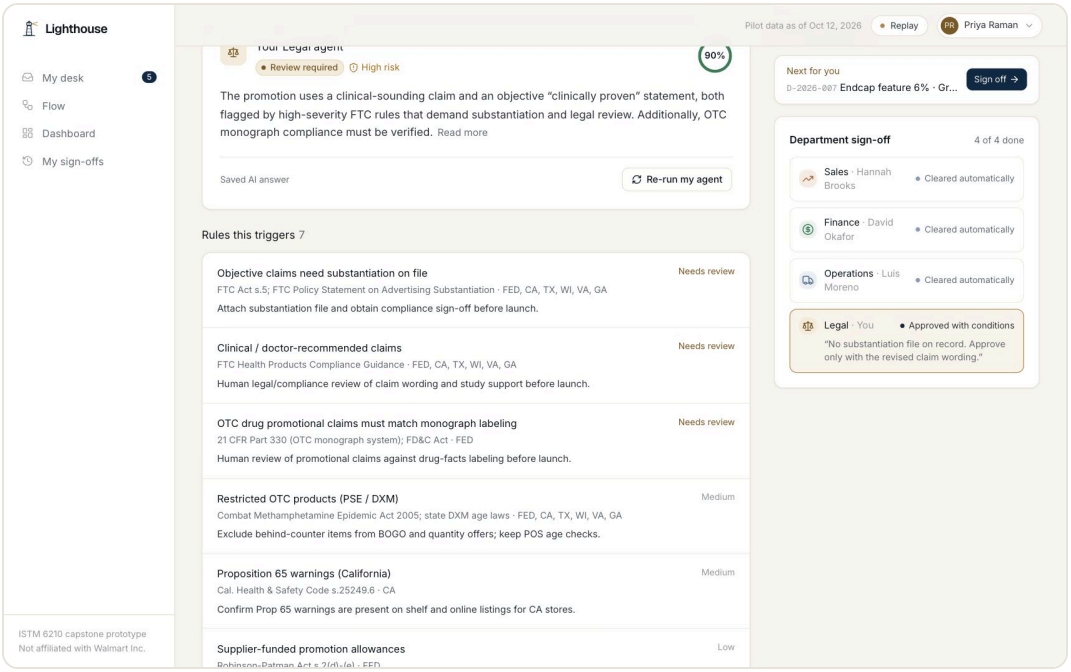


The name menu: the badge is how much is waiting for each person; Presenter tools and Sign out at the bottom.



Priya's lane: claim, findings, rules and her sign-off form.

After signing, a **Next for you** card points to her next case, so nobody hunts for the next step.

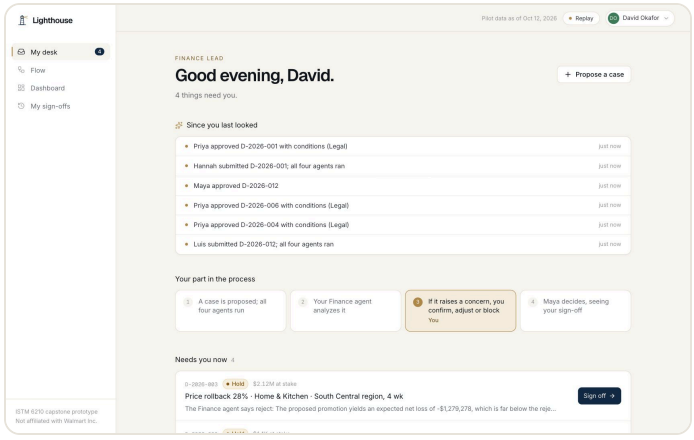


Done: her sign-off, the Next for you card, and the four departments' status.

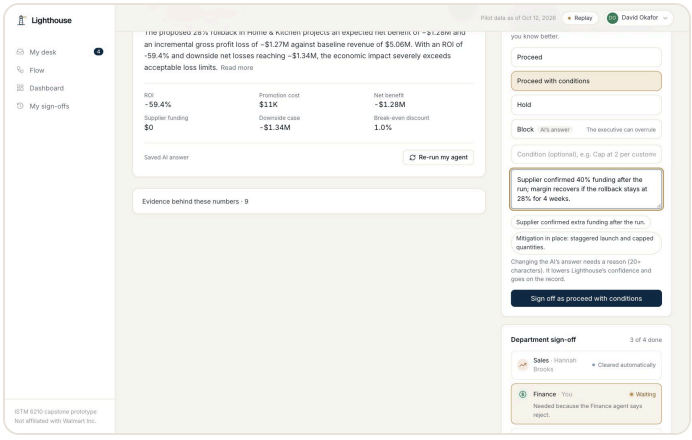
4 **David disagrees with his agent.** DAVID · FINANCE LEAD

Switch to David and open *My desk*: **Since you last looked** lists what just happened (Hannah submitted, Priya approved). Open D-2026-003 (28% Home & Kitchen). His agent says Reject: net benefit –$1.28M. Click **Re-run my agent** to show that he owns it. Then choose **Proceed with conditions**, give the reason and sign.

> *"The AI is advice; the department head is accountable. Changing the agent's answer needs a reason, costs Lighthouse confidence, and both versions stay on the record."*



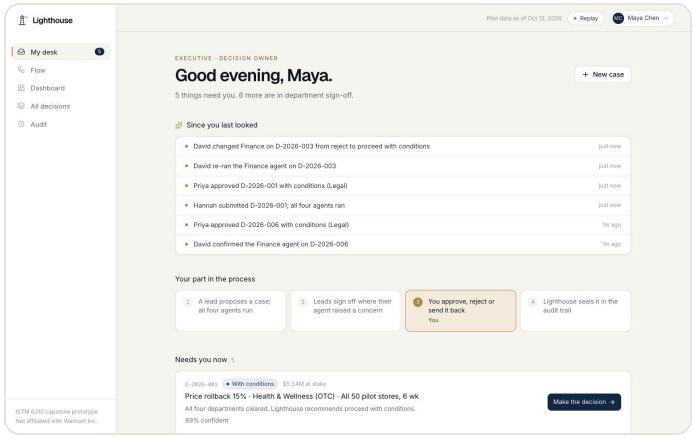
Since you last looked: every hand-off shows up on the next person's desk.



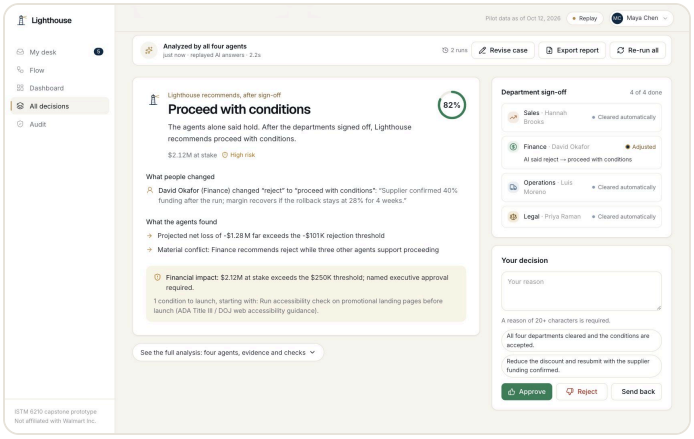
The AI's answer is marked; changing it needs a reason. Tap a suggested reason, then edit it.

5 **Maya's desk.** MAYA · EXECUTIVE

Switch to Maya. Her desk opens with what changed, then the decisions waiting for her. Each row says where the departments stand and how confident Lighthouse is; for D-2026-003 it reads *"AI alone said hold; after sign-off, proceed with conditions"*, before she opens anything.



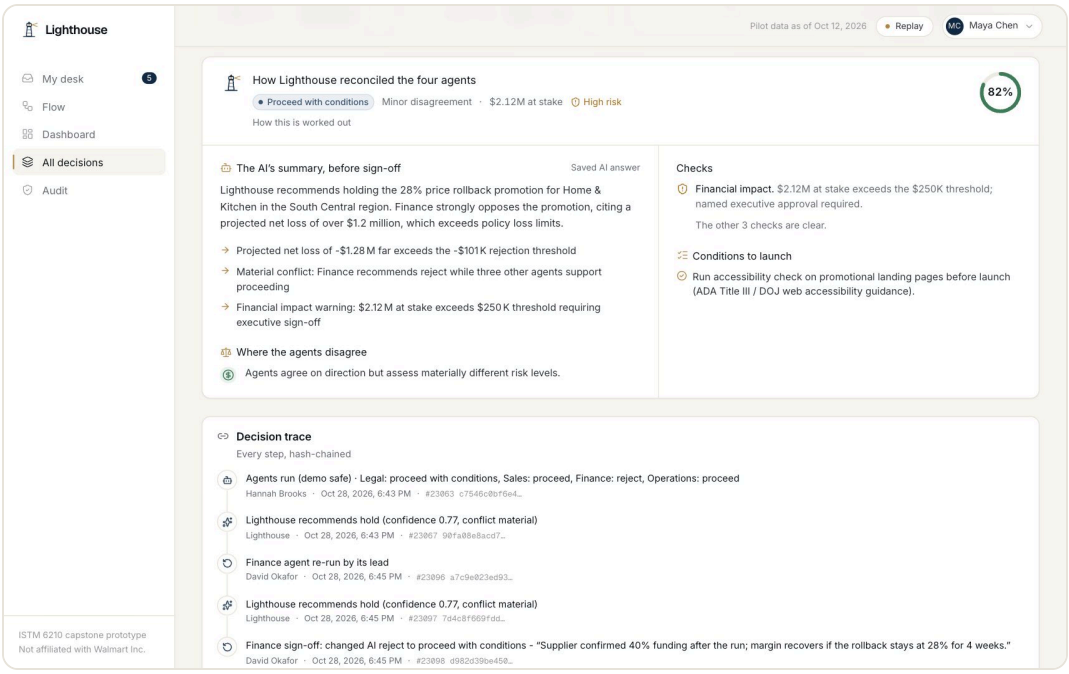
Maya's desk: what changed since she last looked, then her decisions.



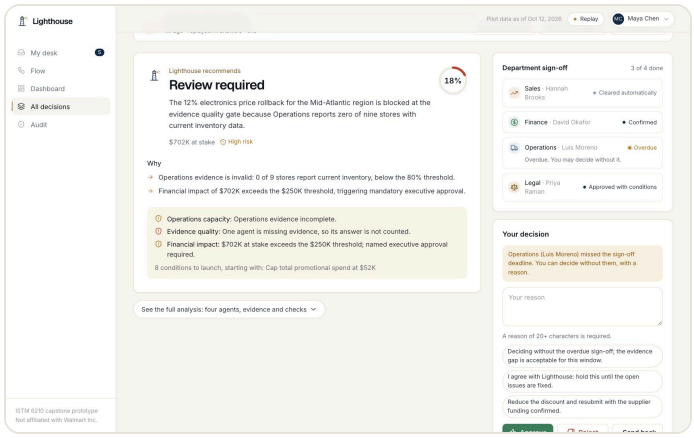
The decision brief on D-2026-003: the AI said hold, David's sign-off changed it.

6 **Human vs AI, side by side.** Open D-2026-003. The brief reads *"Proceed with conditions. The agents alone said hold. After the departments signed off, Lighthouse recommends proceed with conditions."* Under *What people changed* is David's change and his reason, and the sign-off panel shows *AI said reject → proceed with conditions*. Click **See the full analysis** for the four agents, the evidence and every check. Then open D-2026-006 (Operations missed the deadline, so Maya may decide without it, with a reason) and D-2026-005 (a Texas rule prohibits it, so approval is locked for everyone).

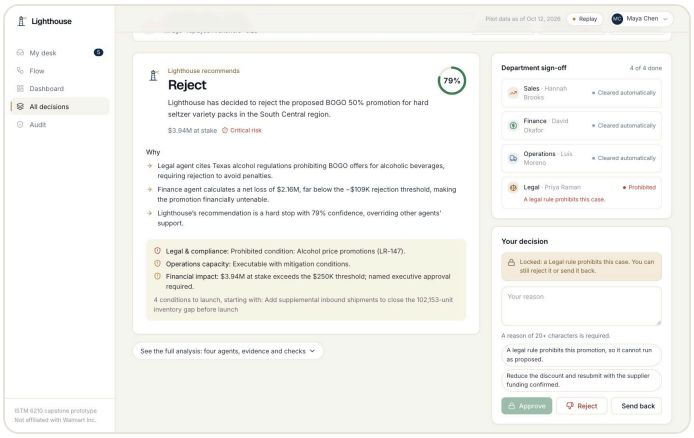
> *"Nothing is averaged away. Maya sees exactly where a person overruled their AI, and why, before she decides."*



See the full analysis: how Lighthouse reconciled the four agents, the open checks and the decision trace.

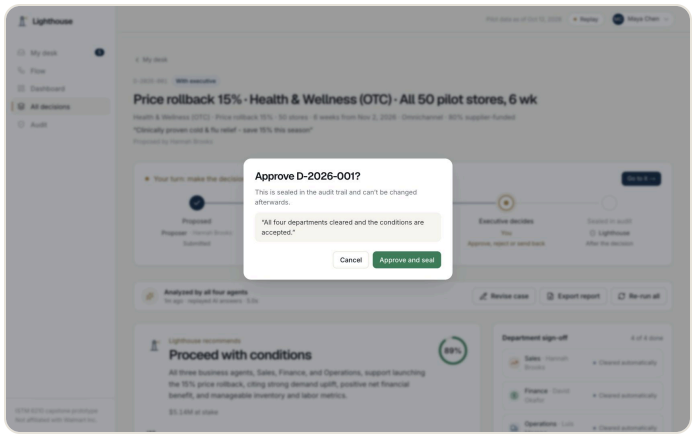


D-2026-006: Operations is overdue, so Maya may decide without it. Legal never escalates.

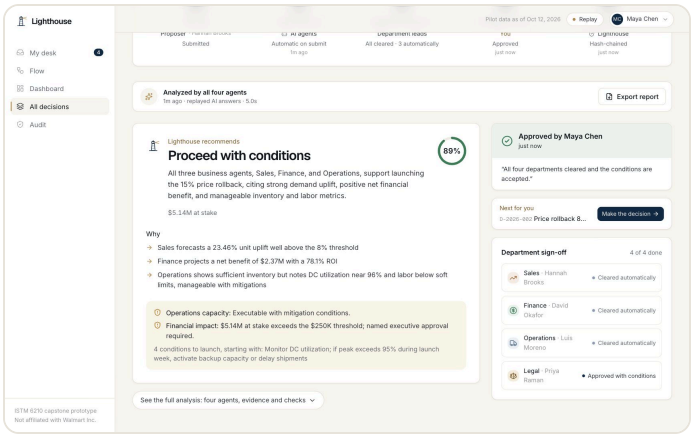


D-2026-005: prohibited by a legal rule, so approval is locked.

7 **Approve D-2026-001.** Open it, tap the suggested reason *“All four departments cleared...”* (or type your own) and click **Approve**. Because a decision is final, Lighthouse asks once: *“This is sealed in the audit trail and can’t be changed afterwards.”* Click **Approve and seal**. The journey bar reads *Decided and sealed in the audit trail*, and the **Next for you** card points to her next decision.



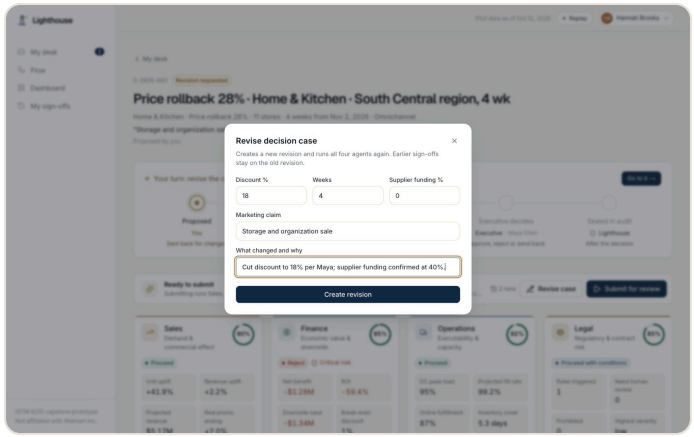
One confirmation before a decision is sealed. Send back needs none.



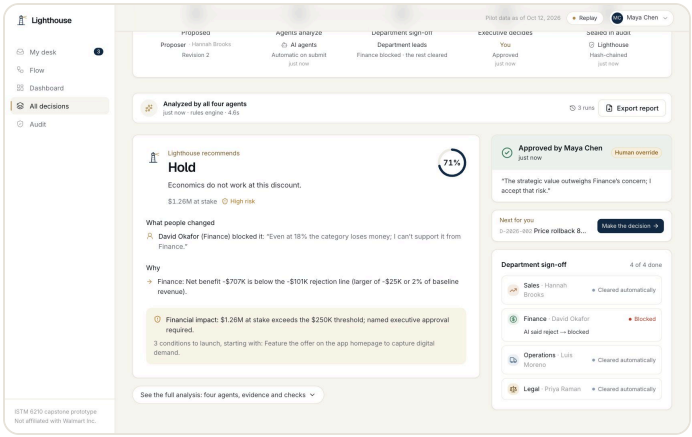
The decision on record, with the Next for you card.

8 **Optional: the full loop.** IF YOU HAVE 90 SECONDS

On D-2026-003 Maya clicks **Send back** (“Cut the discount to 18%…”). Hannah’s desk now says *Revise the case*, with Maya’s words; she clicks **Revise case**, sets 18%, and the four agents re-run by themselves. David’s agent still says reject, so he clicks **Confirm the agent: block it** and gives a reason. Maya’s desk says *Finance blocked it. You can overrule with a reason*; the brief names who blocked it, she taps the overrule reason, approves and seals, and it is recorded as a **Human override**.

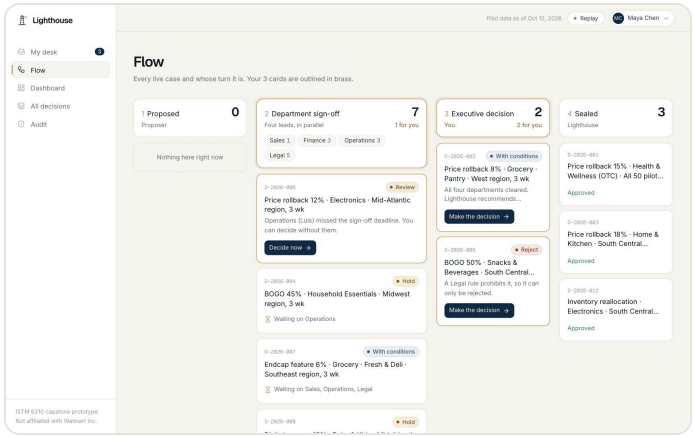


Hannah revises: a new revision, and the agents re-run automatically.

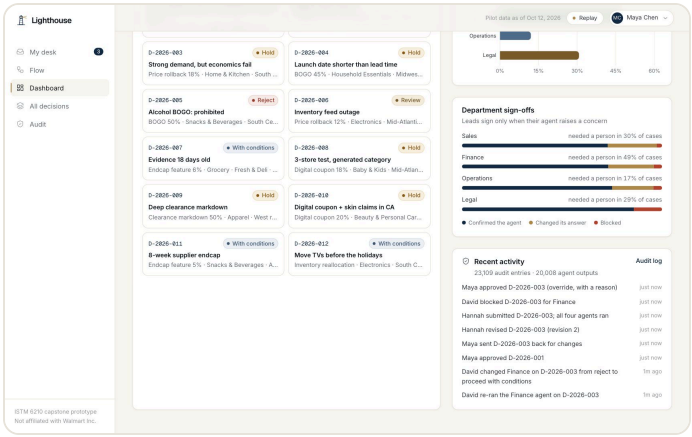


Maya overrules Finance’s block; her reason and the override are on record.

9 **Flow and Dashboard.** *Flow* shows every live case by stage. Each column says how many cases are there and how many are yours; the sign-off column also counts per department, and your cards are outlined in brass. The *Dashboard* covers all 5,000 decisions, including how often each department needed a person and how often they agreed with their agent.



Flow: four stages, whose turn it is, and how many are yours.

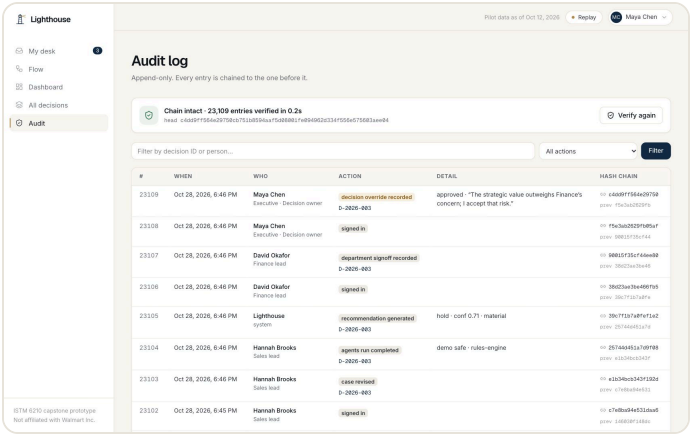


Dashboard: department sign-offs (confirmed, changed, blocked) and recent activity.

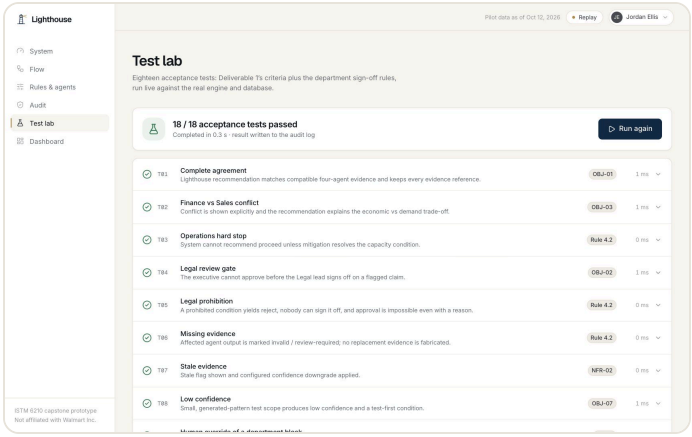
10 Audit and tests. MAYA, THEN JORDAN

Audit → Verify chain re-hashes about 22,000 entries in under a second. Then switch to Jordan (PIN 4210) and open *Test lab → Run all tests*: 18 acceptance tests run live, including the sign-off rules.

> *"If anyone edited a past decision or sign-off, verification would point to the exact entry that changed."*

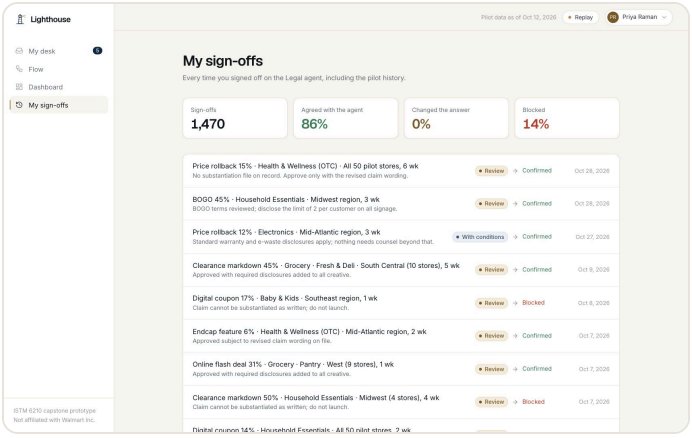


Tamper-evidence check over the full audit chain.

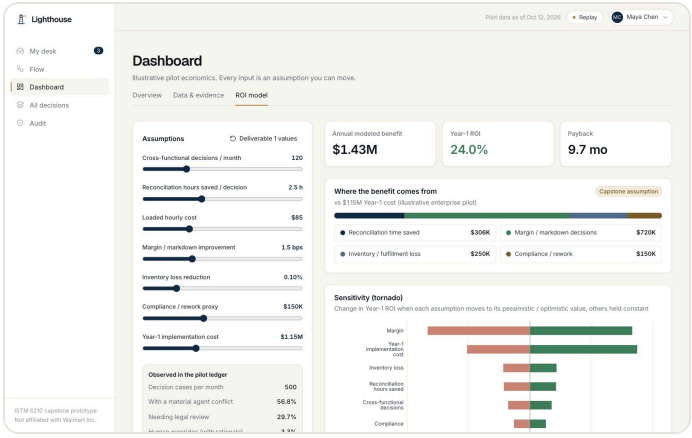


18 of 18 acceptance tests passing against the real engine and database.

If you have extra time: as Priya, open *My sign-offs* (how often she agreed with her agent); open *Dashboard → Data* and *ROI*.



My sign-offs: each lead's record of confirming, changing or blocking their agent.



ROI model (Deliverable 1: $1.43M benefit, 24% ROI, 9.7-month payback) with live sliders.

6 · REFERENCE

The twelve demo cases after a reset

With Replay on, outcomes are exactly as listed. Live AI wording varies a little from run to run, but the governance rules fix the outcome.

CASE	SCENARIO	STARTING STATE	WHO MUST SIGN · WHAT IT SHOWS
D-2026-001	15% OTC rollback, “clinically proven” claim	Proposed (submit live)	Legal only. The anchor case: legal gate, then approval.
D-2026-002	8% pantry rollback, supplier-funded	With executive	Nobody: all four cleared automatically. With conditions
D-2026-003	28% Home & Kitchen rollback	In sign-off	Finance. Demand strong, economics fail: Hold. Best case for a human adjustment.
D-2026-004	BOGO launching in 3 days	In sign-off	Operations (Legal already signed). Lead time beats launch: hard stop, Hold.
D-2026-005	Hard seltzer BOGO in Texas	With executive	Nobody: a legal rule prohibits it. Reject, approval impossible.
D-2026-006	Electronics during an inventory feed outage	In sign-off, Ops overdue	Operations is 54h overdue, so it escalates to Maya. Evidence invalid, never imputed.
D-2026-007	Deli endcap on 18-day-old evidence	In sign-off	Sales, Operations, Legal: stale data drops every agent below 65% confidence.
D-2026-008	3-store baby digital coupon	In sign-off	Finance (hold) and Legal (children’s data).
D-2026-009	50% apparel clearance	In sign-off	Finance (reject) and Legal (reference-price rules).
D-2026-010	Beauty coupon, skin-repair claim, California	In sign-off	Finance and Legal (privacy plus cosmetic-drug claim).
D-2026-011	8-week supplier endcap, all stores	In sign-off	Legal (exclusivity terms).
D-2026-012	Move 6,000 TVs before the holidays	Approved	Proposed by Luis (Operations); shows a sealed case.

Desk counts after a reset: Maya 3 · Hannah 2 · David 4 · Luis 3 · Priya 5.

7 · RELIABILITY

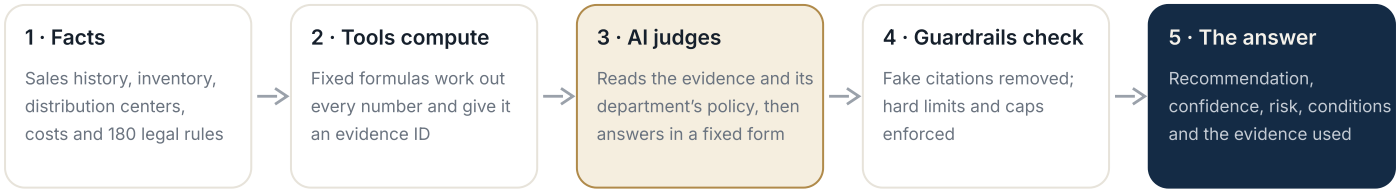
Live AI, Replay and the backup plan

LAYER	WHAT IT IS	WHEN IT IS USED
Gemini (free)	The newest Gemini Flash model the key can use	First choice on the deployed site
Groq (free)	gpt-oss-120b → Qwen → gpt-oss-20b	Automatically if Gemini errors or times out (15s). A failed provider is skipped for 10 minutes.
Rules engine	Deterministic version of each agent	If no AI answers; cards are labelled “Rules engine”
Replay	The saved AI answers for each demo case	Name menu (top right) → Presenter, when the network is unreliable

GWU campus network: campus Wi-Fi blocks Google’s Gemini address. The deployed site isn’t affected (Vercel’s servers call Gemini, not your laptop). Running locally on campus, the app falls back to Groq.

What happens when a case is submitted

Submitting a case starts four agents at the same time: Sales, Finance, Operations and Legal. Each one goes through the same five steps. The key idea: **the computer does the maths, the AI does the judging, and the rules have the last word.**



Why split the work this way

AI models are good at weighing evidence and explaining it, but they can get arithmetic wrong or invent facts. So in Lighthouse the AI never calculates anything. Every number (demand, profit, warehouse load, which rules apply) is worked out first by fixed formulas called **tools**, and each result gets an **evidence ID** such as `EV-…-F3`. The AI reads that list and must cite the IDs it relied on.

What every agent hands back

Recommendation	Proceed, Proceed with conditions, Hold, Reject or Review required
Confidence	0–100%: how sure it is, given how good its evidence is
Risk	Low, Medium, High or Critical
Conditions	What must be true to go ahead (for example "cap spend at $52K")
Evidence	The IDs of the facts it used
Rationale	Two to four plain sentences for a busy executive

The guardrails, in plain words

- **No made-up sources.** A citation to evidence that doesn't exist is deleted. If nothing valid is left, the answer isn't counted.
- **Hard limits win.** If a legal rule prohibits the case, the Legal agent must say Reject. If the warehouse would overflow, Operations must say Hold. The AI can't talk its way around them.
- **Risk can't be played down.** It is raised to at least what the evidence shows (for example, the most serious legal rule triggered).
- **Confidence is capped when data is weak.** Old or missing data puts a ceiling on how sure an agent may be (see the next pages).
- **Case text is treated as data.** Anything typed into a proposal is quoted, never followed as an instruction.

If the AI is unavailable

Lighthouse asks Gemini first, then Groq. If neither answers, a **rules engine** makes the call with the same tools and fixed rules, and the card says so. **Replay** reuses saved AI answers so a demo gives identical results every time.

The four agents

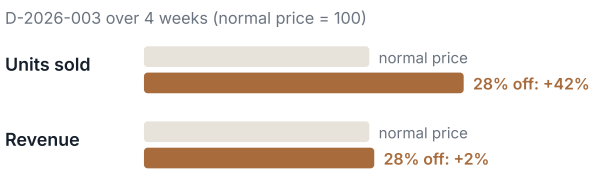
Each agent answers one question for its department and nothing else. The examples use **D-2026-003**: a 28% price cut on Home & Kitchen in 11 stores for 4 weeks.

Sales · “Will customers buy more?”

Looks at: the last 56 days of sales in those stores, the season for the exact dates, how strongly the category reacts to price (its *elasticity*), and real past promotion days in the same stores.

How it forecasts: a price cut sells more units, and the mechanic itself adds a little (an endcap display adds 12%). That estimate is blended 75/25 with what real past promotions did, and trimmed for promotions longer than 4 weeks, when shoppers tire of the deal.

Decides: under 3% more units → Hold · 3–8% → Proceed with conditions · 8% or more → Proceed. Too little sales data → Review required.



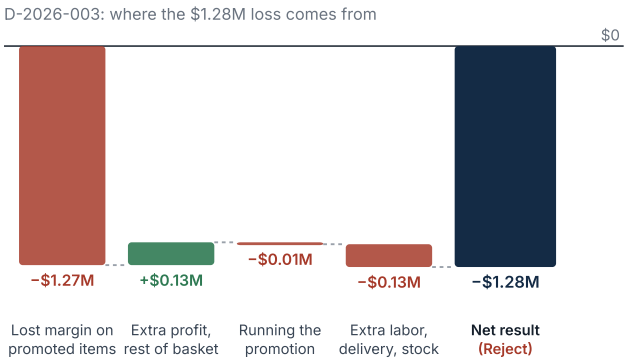
The cut sells 42% more units, but each sells for 28% less, so revenue rises only 2%. Sales says Proceed; it only judges demand.

Finance · “Do we make money?”

Looks at: the Sales forecast, the category’s margin (35% here: $35 profit on every $100 sold at full price), supplier funding, and the cost of running the promotion.

How it works it out: extra profit from the extra units, plus profit on other items shoppers add to the basket, plus supplier funding, minus the discount given away, labor, delivery and the cost of holding stock. It also tests a bad case (40% fewer extra sales) and finds the largest discount that still breaks even.

Decides: a loss bigger than $25K or 2% of revenue (whichever is larger) → Reject · a smaller loss → Hold · profit, but a loss in the bad case → Proceed with conditions · profit in both → Proceed.



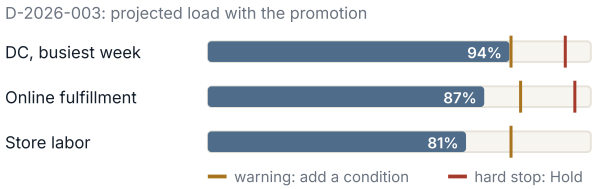
At 28% off, the profit on every $100 of goods (at full price) falls from $35 to $7, so 42% more units can’t make up for it. Only a 1% discount breaks even. Finance says Reject.

Operations · “Can we actually do it?”

Looks at: stock on hand and on the way, safety stock, how busy the distribution center (DC) usually is on those dates, store labor, online order capacity and how long restocking takes.

How it works it out: adds the forecast extra units to the normal load, with launch week weighted heavier because demand is front-loaded. It checks whether there is enough stock to open, whether a restock can arrive in time, and what share of orders would be filled (the *fill rate*).

Decides: Hold (a hard stop) if the DC passes 112%, online orders pass 115%, the fill rate falls under 92%, or restocking can’t arrive before launch. Past a warning line (DC 95%, online 98%, labor 95%) → Proceed with conditions, with a fix such as “stage deliveries by week”. Fewer than 80% of stores reporting inventory → Review required.



Everything sits under the warning lines and 99% of orders would be filled. Operations says Proceed.

Legal · “Are we allowed to?”

Looks at: a catalog of 180 active rules (federal and state), each with a trigger: category, promotion type, channel, state, minimum discount, length, flags such as “alcohol” or “children’s data”, and words in the marketing claim such as “clinically proven”.

How it works it out: a rule applies only when *every* trigger it lists matches the case. This is exact matching, not AI guessing, so Legal is rarely unsure. The AI then explains the matches and their conditions in plain words.

Decides:

A matched rule **prohibits** it	Reject, risk Critical
A matched rule needs a **lawyer’s review**	Review required
Only informational rules match	With conditions
No rule matches	Proceed

Risk is at least as high as the most serious rule matched. Because matching is exact, the rules engine gives Legal 90–97% confidence. Outputs are compliance-risk flags, not legal advice.

One low-severity Texas rule matches (accessible web content for the online offer). Legal says Proceed with conditions.

How sure is each agent?

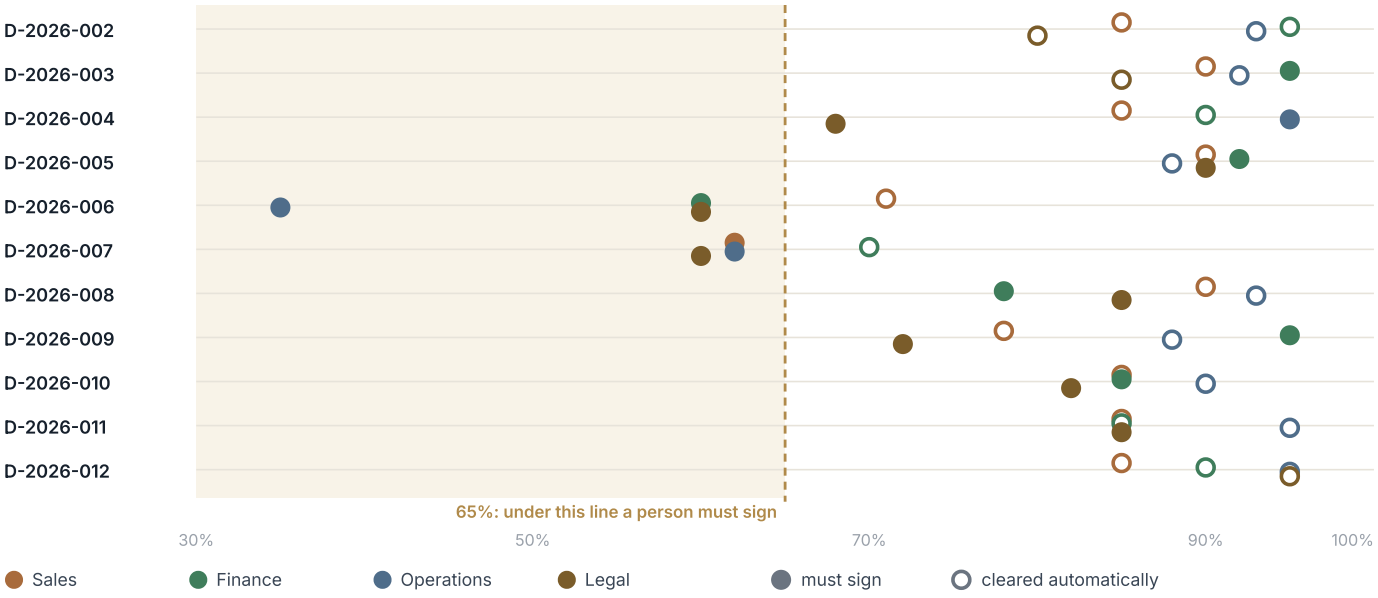
Each agent's confidence answers: *"How sure am I, given how good my evidence is?"* The AI sets it from 0% to 100% and is told that stale or incomplete evidence must lower it. Then guardrails put a ceiling on it:

WHEN THIS HAPPENS…	…THE CEILING IS
Sales data is more than 7 days old	Sales and Finance at most 72%
Store inventory data is more than 5 days old	Operations at most 70%
Under 60% of the expected sales records exist, or there is no sales baseline	Sales or Finance at most 50%, and the answer isn't counted
Under 80% of stores report current inventory	Operations at most 45%, and the answer isn't counted

Without AI, the rules engine starts each agent at a fixed level and takes points off for weak evidence. Sales starts at 88%, minus 12 for fewer than 30 past promotion days to learn from, 10 for a test in under 5 stores, and 12 for stale data. Finance and Operations start at 86%, losing points for stale data, and when the result sits right at break-even or near a capacity limit. Every agent stays between 30% and 95%.

Why it matters: confidence decides who has to sign

A department lead only signs when their agent raises a concern: it says Hold, Reject or Review required, it is **under 65% confident**, or its evidence is missing. Everything else clears automatically. The chart shows every agent's confidence on the demo cases.



D-2026-007: all four agents say proceed, but its data is 18 days old, so three of them sit under 65% and three leads must sign. D-2026-012: everyone is sure and nobody objects, so nobody has to sign at all.

8 · HOW IT WORKS

From four answers to one confidence score

Lighthouse confidence says how much to trust the combined recommendation. It is **not an average**. It is four factors multiplied together, so one weak link pulls the whole score down. If only half the data exists, the score halves, however sure the four agents sound. An average would hide that.

Agents' confidence	×	Evidence complete	×	Evidence fresh	×	Agreement	=	Lighthouse confidence
weighted by department		share of data present		−12 points per stale source		1 − disagreement penalty		
91%		100%		100%		85%		77%

Brass figures: D-2026-003 before sign-off. Result kept between 15% and 95%.

Factor 1 · The agents' confidence, weighted by department

Each agent's confidence is multiplied by its department's weight, and the four results are added up.



Finance weighs most because money is the most common deciding factor. Legal weighs least because it doesn't need weight: a legal prohibition is a veto, whatever the other three say.

Factor 2 · Is the evidence complete?

The average of two shares: how many expected sales records exist, and how many stores report current inventory. Then 15 points come off for every agent whose answer isn't counted. It never goes below 30%.

D-2026-006: sales records 100%, stores reporting 0 of 9 → (100% + 0%) ÷ 2 = 50%, minus 15 for Operations = **42.5%**.

Factor 3 · Is the evidence fresh?

Starts at 100% and loses 12 points for each source past its age limit: sales data (7 days) and store inventory (5 days). So 100%, 88% or 76%.

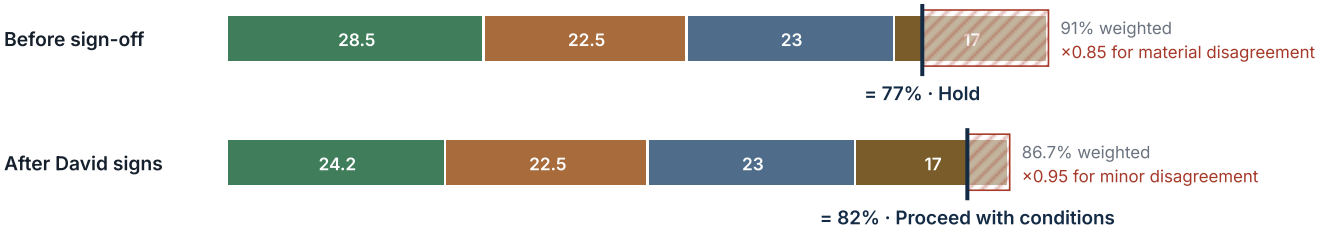
Factor 4 · Do the agents agree?

DISAGREEMENT	WHEN	TAKEN OFF
None	All point the same way	0%
Minor	Same direction, but Legal needs a review, or risk levels are far apart (e.g. Low vs High)	5%
Material	Some support and some oppose, or an answer is missing evidence	15%
Hard stop	Legal prohibits or Operations can't execute	8%, then 5% more

A hard stop costs less than a real split because the answer is clear: the case can't go ahead as proposed. The final score always stays between 15% and 95%; Lighthouse never claims certainty.

A worked example, and what a person's sign-off changes

D-2026-003 before and after David (Finance) changes his agent's Reject to Proceed with conditions. These are the exact figures the app shows.



Each block is that department's confidence × its weight (points out of 100). Hatched = taken off for disagreement.

Before sign-off: 77%, Hold

1. Weighted confidence: Finance 95% × 0.30 = 28.5, Sales 90% × 0.25 = 22.5, Operations 92% × 0.25 = 23.0, Legal 85% × 0.20 = 17.0. Total **91%**.
2. Evidence complete (100%) and fresh (100%): nothing comes off.
3. Finance opposes while three support: material disagreement, × 0.85 → **77%**.
4. Finance says Reject and Sales says Proceed, so the recommendation is **Hold**.

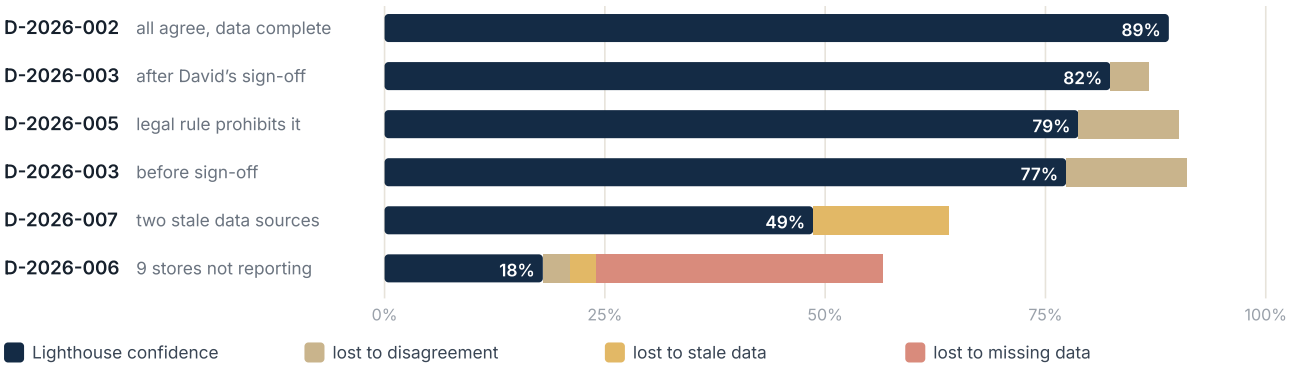
After David signs: 82%, Proceed with conditions

1. David's position replaces the agent's: Proceed with conditions, with his reason on record.
2. A changed answer costs confidence: Finance's 95% × 0.85 = 81%, so its share drops from 28.5 to 24.2. Total **86.7%**.
3. Nobody opposes now, but risk levels still differ (Finance High, the rest Low): minor disagreement, × 0.95 → **82%**.

Confidence goes up because an accountable person resolved the disagreement. The 15% cost means a human change is never free, and the report still says "the agents alone said hold".

Five demo cases: where confidence is lost

Each bar starts at the weighted agents' confidence. The coloured parts show what the other three factors took away.



How the recommendation is chosen

The recommendation is **not** a vote and **not** an average. Lighthouse checks these questions in order, and the first one that is true decides. Safety and legality come first, then money, then agreement.

	IF THIS IS TRUE…	LIGHTHOUSE RECOMMENDS	NOTE
1	A legal rule prohibits it	Reject	Nobody can approve it.
2	Any agent is missing required evidence	Review required	Missing data is never guessed.
3	Operations can't execute it (hard stop)	Hold	e.g. the warehouse would pass 112%.
4	Finance says reject	Hold or Reject	Hold if Sales says proceed, else Reject.
5	Finance says hold	Hold	The case waits until the economics work.
6	Two or more agents say hold or reject	Hold	
7	Exactly one agent says hold or reject	Proceed with conditions	Its concern becomes a condition.
8	Legal needs a lawyer to review it	Proceed with conditions	Legal's sign-off is required before launch.
9	Any agent attaches conditions	Proceed with conditions	
10	None of the above	Proceed	All four support it.

Everything else in the report

- **Risk** is the highest risk any of the four agents reported. A legal prohibition makes it Critical.
- **Conditions** from all four agents are merged, with duplicates removed (up to 8).
- **Checks** flag what needs attention, such as over $250K at stake, which needs the named executive.

After the departments sign, it all runs again

- Each signed department's position **replaces** its agent's answer.
- **Confirmed**: nothing changes. **Changed**: that agent's confidence × 0.85. **Blocked**: the department now opposes it.
- A lead who signs a positive position takes ownership of the risk, so it is capped (Legal at Medium, others at High). A Legal rejection makes it Critical.
- The same questions and the same formula run on the signed positions, and the report keeps both results: *"The agents alone said hold. After the departments signed off, Lighthouse recommends proceed with conditions."*

In one sentence: tools work out the numbers, each AI agent judges its own department's evidence, guardrails keep it honest, Lighthouse combines the four with fixed rules and a confidence score that punishes weak links, the department leads own their agents' answers, and the executive makes the call.

9 · QUESTIONS

Likely questions

QUESTION	ANSWER
Why doesn’t every lead review every case?	Review fatigue. Leads sign only where their agent has a concern; clean answers clear automatically. Every lane still shows who owns it, and any lead can step in.
Isn’t the human just rubber-stamping the AI?	The record says otherwise. Changing an answer needs a reason and is shown side by side with the AI; the Dashboard tracks how often each department confirmed, changed or blocked its agent.
What if a lead never responds?	After 48 hours a Sales, Finance or Ops lane escalates and the executive may decide without it, with a reason. Legal never escalates.
Can the executive overrule Legal?	No. A Legal rejection or a prohibiting rule locks approval for everyone.
Why is the administrator separate?	Separation of duties: the person who sets the rules and thresholds can’t also decide cases under them.
Is this real Walmart data?	Demand patterns are real public Walmart data from Kaggle (M5 daily sales; 45-store weekly sales), calibrated to the FY2026 10-K. Stores, inventory, rules and costs are generated and labelled. No Walmart internal data.
What if the AI hallucinates?	Numbers come from tools; the AI must cite evidence IDs; unknown IDs are removed; hard limits apply after the AI answers (test T12). The department lead is the last check.
Prompt injection?	User text is passed as quoted data, screened for injection patterns, and never treated as instructions (test T13).
Does it meet the 60-second target?	Live runs take about 6–15 seconds; a single-agent re-run about 3–5; reconciliation takes milliseconds (test T14).

10 · HELP

Troubleshooting

- **A lane says overdue:** only D-2026-006 should (it is set up that way). If others do, reset the demo from the name menu.
- **An agent card says “Rules engine”:** no AI provider answered. The result is still valid. Re-run, or switch to Replay.
- **Approve is locked:** by design. The panel says which department is still open; switch to that person.
- **A demo case is already decided:** name menu (top right) → Presenter → *Reset demo cases*.
- **Need a printout:** on any case, *Export report* → Print / Save as PDF. It lists every department’s sign-off.